Test Report:

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Module: MSO2700 Software Design

Task: Tree Sort Implementation and Testing

Purpose:

The purpose of this testing is to verify that the tree sort method correctly sorts a list of integers into ascending order.

Required Test Case:

Input: (10, 5, 8, 3, 1, 5)

Expected Output: (1, 3, 5, 5, 8, 10)

Result: The method returned the correct sorted list.

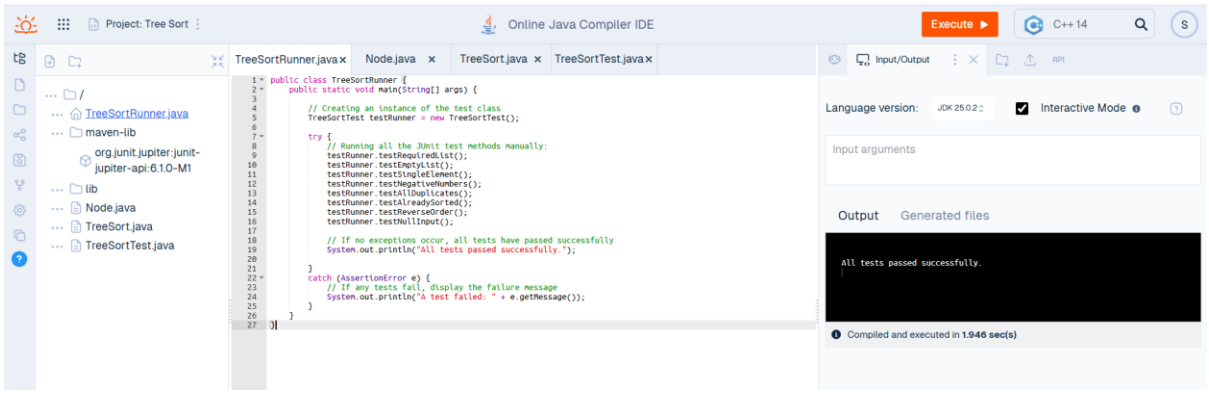
Additional Test Cases:

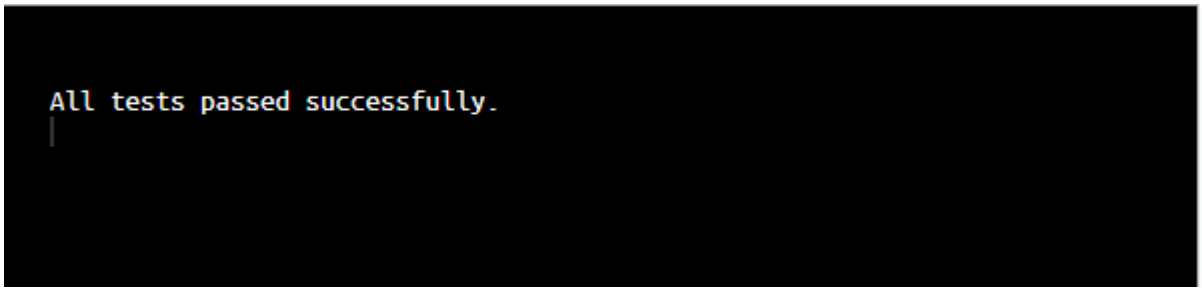
The following additional tests were carried out to ensure robustness:

- Empty list → returns empty list
- Single element → unchanged
- Negative numbers → correctly sorted
- Duplicate values → handled correctly
- Already sorted list → unchanged
- Reverse order → correctly sorted
- Null input → handled safely

All tests passed successfully.

Test Evidence:





Conclusion:

The tree sort implementation works correctly for the required input and handles a variety of edge cases successfully.